Partisan Vision? Partisan Bias in Simple Visual Evaluations

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October 29, 2026

Abstract

In the current era of partisan polarization, partisanship strongly colors partisans' worldviews. But does it cause partisans to *see* different things? We test the hypothesis using two survey experiments and a survey. In the experiments, respondents count errors in a short passage or poorly parked cars in a photograph. We randomly vary whether the passage's author or the people who parked the cars are described as Democrats or Republicans. We also replicate the writing experiment on Lucid. Democrats count an average of 8.7 poorly parked cars outside a Republican Party meeting, compared with 6.1 outside a Democratic Party meeting, despite seeing the same photograph. Republicans' corresponding difference is small and imprecisely estimated. Both parties find slightly more writing errors when the author belongs to the opposing party, but the confidence intervals include zero in both studies. In a separate survey, respondents count people wearing masks in a 10-second video. Republicans count 0.87 more than Democrats (95% confidence interval: 0.41 to 1.33) when responses are capped at 15. Without the cap, extreme responses reverse the difference. The experimental estimates consistently point toward partisan bias, with the clearest individual effect in Democrats' judgments of parking errors.

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Partisans are increasingly polarized (Iyengar, Sood and Lelkes 2012), with partisan cleavages outstripping some of the longer-standing racial cleavages (Iyengar and Westwood 2015). In this paper, we explore whether polarization affects how partisans "see." We test the hypothesis with simple visual evaluative tasks. In particular, we field two survey experiments and a survey. Across both writing studies, Democrats and Republicans find more errors in text attributed to the opposing party. The clearest individual effect is in Democrats' judgments of parking errors.

Group loyalties can shape how people judge the same scene. In the classic study of a contentious football game, Princeton and Dartmouth students differed in the infractions they attributed to the teams (Hastorf and Cantril 1954). In a political setting, changing the purported cause of a protest changed judgments of whether the same demonstrators were obstructing or threatening pedestrians (Kahan et al. 2012). Related experiments find that motivations influence interpretations of ambiguous visual stimuli (Balcetis and Dunning 2006) and that political agreement affects which photographs people regard as representative of a candidate's skin tone (Caruso, Mead and Balcetis 2009).

These findings fit a broader literature on motivated political reasoning. People evaluate congenial arguments more favorably and scrutinize contrary arguments more closely (Taber and Lodge 2006). Party identification can organize attention, memory, and evaluation around defending one's political group (Van Bavel and Pereira 2018). A meta-analysis of experiments that hold information constant while varying its political implications finds bias among both liberals and conservatives (Ditto et al. 2019).

But partisan differences in survey answers need not be large or uniform. Accuracy incentives reduce partisan gaps in answers to factual questions (Bullock et al. 2015; Prior, Sood and Khanna 2015), and vague response options can inflate their size (Roush and Sood 2023). Even a straightforward comparison of inauguration photographs can elicit answers favoring a respondent's preferred candidate (Schaffner and Luks 2018). Our experiments ask

whether simply attributing a piece of text or a parking lot to a party changes the number of errors people find. Respondents assess the material in front of them and give a count, rather than recall a political fact or choose among verbal descriptions of it.

1 Data and Research Design

To assess how partisans *see*, we fielded two survey experiments with respondents recruited by YouGov as part of a Cooperative Congressional Election Study (CCES) module. In the first experiment, we presented people with a short passage and asked them to count the number of errors in it. We manipulated the perceived party of the person writing the text (see Figure SI 1.1). In the second experiment, we showed people a photo of a parking lot and asked them to estimate the number of poorly parked cars. We manipulated the perceived party of the people who parked the cars by changing the caption indicating where the photo was taken (see Figure SI 1.2). We replicated the first experiment on Lucid, an online survey platform (Coppock and McClellan 2019). The Lucid version uses a shorter passage.

We complemented the survey experiments with a partisan evaluative task in a survey. In a survey conducted on MTurk, we asked respondents to watch a short video and estimate how many people in the video were wearing masks. In particular, our directions were as follows: "Please watch the following short (10-second) video. You will be asked a question about it on the next screen. ... How many people in the video were wearing masks?"

We analyze the unmatched CCES sample without survey weights and report results from the matched, weighted sample in the replication materials. The writing task appeared before the election, and the parking task afterward. The main results focus on Democrats and Republicans, including leaners. The appendix reports descriptive results for all groups, including independents. For Lucid, we exclude previews and retain respondents who consented and passed the attention check. For MTurk, we pool the two survey waves and retain

respondents who said they gave humorous or insincere answers never or rarely. The original instruments, party coding, sample counts, and sensitivity analyses are included in the replication materials.

Figure 1 presents mean error counts alongside differences between the out-party and in-party conditions. Intervals for the means use the t distribution; intervals for the differences use Welch's method. All intervals are 95% confidence intervals. We examine uncapped and capped responses, the influence of individual observations, and differences across the two MTurk waves. The CCES analysis description on OSF was registered in November 2021, after the 2020 fieldwork (https://osf.io/rqj8u).

2 Results

Across both writing studies, Democrats and Republicans find more errors when the text is attributed to the opposing party (Figure 1). The four estimates consistently point toward partisan bias, although each confidence interval includes zero.

As Appendix Table SI 2.1 shows, Democrats find 8.7 mistakes when they think the text is written by a Democrat compared to 8.9 mistakes when they think the text is written by a Republican. On the other hand, Republicans find 7.6 mistakes on average when they think the text is written by a Democrat and 7.1 mistakes when they think it is written by a Republican. The estimated differences are small. The point is especially clear when you look at the medians, which are the same within each party. The confidence intervals for the differences include zero, but also leave room for larger effects (Appendix Table SI 2.5).

Appendix Table SI 2.2 presents the results of the replication on Lucid. Democrats find 5.5 (0.6) errors on average when they think Democrats wrote the text vs. 5.9 (0.8) when they think Republicans wrote it. On the Republican side, the corresponding numbers are 5.6 (0.9) (when they think Democrats wrote the text) vs. 4.9 (0.3) (when they think

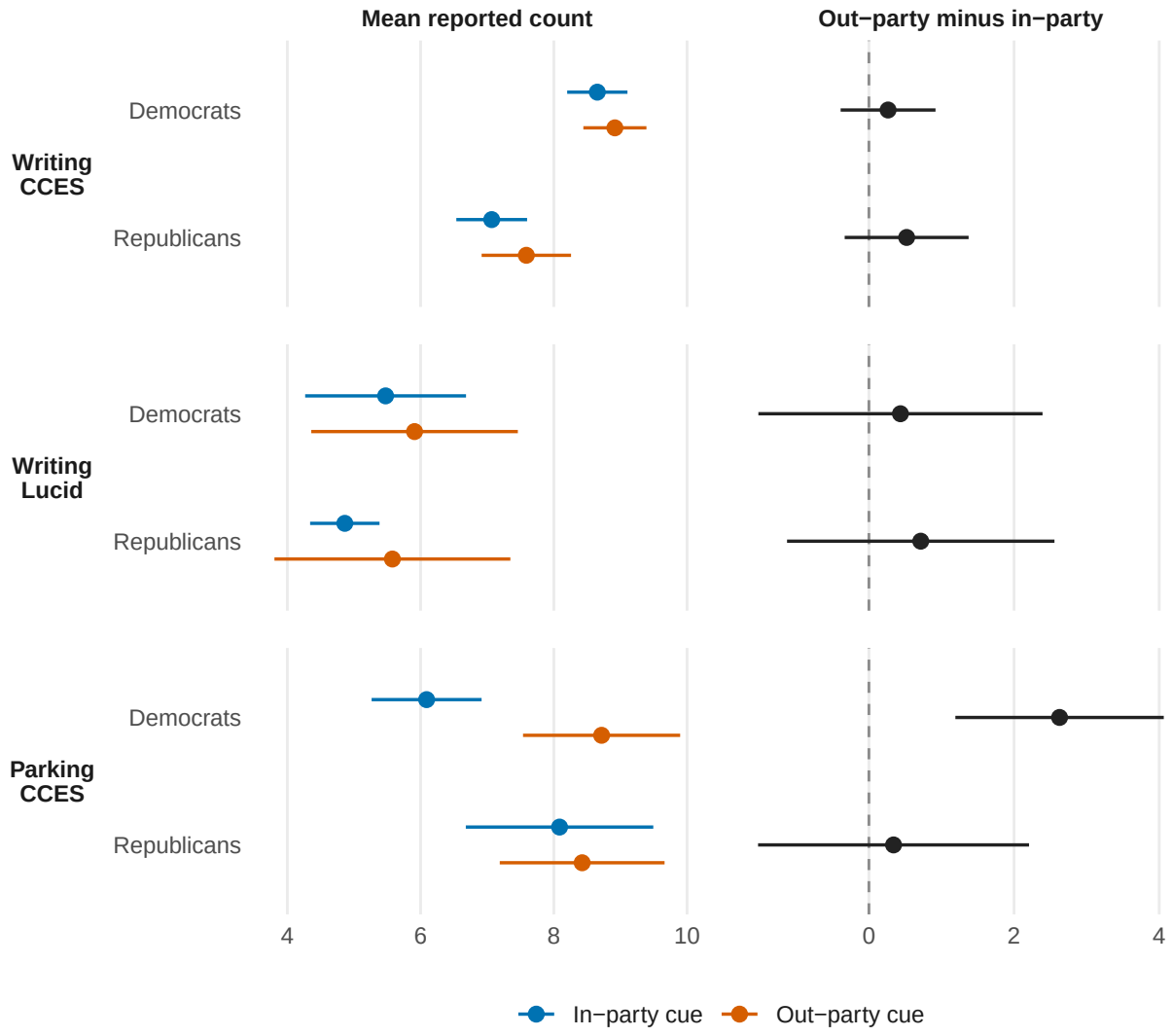


Figure 1: Party cues and error counts. Respondents are Democrats and Republicans, including leaners. Left panels show mean reported counts under in-party and out-party cues; right panels show the out-party minus in-party differences. Positive differences indicate more errors attributed to the opposing party. Counts refer to errors in a passage or poorly parked cars in a photograph. Horizontal intervals are pointwise 95% confidence intervals, using the t distribution for means and Welch's method for differences. CCES results use the unmatched, unweighted sample; Lucid results exclude previews and retain respondents who consented and passed the attention check. The dashed line marks no difference.

Republicans wrote it). Standard errors are in parentheses. These differences are imprecisely estimated (Appendix Table SI 2.5).

We see a larger partisan bias in the experiment that manipulates the perceived party of the people who parked the cars. Democrats on average believe there are 6.1 poorly parked cars outside the Democratic Party meeting while they think there are 8.7 poorly parked cars in front of the Republican Party meeting (see Appendix Table SI 2.3). As above, the medians are much closer at 5.0 and 6.0 for the Democratic Party meeting and the Republican Party meeting, respectively. The mean difference remains positive when any single respondent is omitted. Switching to Republicans, the gap is much narrower—Republicans on average think that there are 8.4 poorly parked cars in front of the Democratic Party meeting and 8.1 in front of the Republican Party meeting. The gap in medians is one.

Turning to the results from the MTurk survey, Republicans count more people wearing masks than Democrats (see Appendix Table SI 2.4). We winsorized the responses at 15 because of extreme responses. Democrats on average spot 3.5 (0.1) masks and Republicans spot 4.4 (0.2) masks. The difference is 0.87 masks, with a 95% confidence interval from 0.41 to 1.33. Republicans also count more masks when responses are capped at 10 or 20. Without a cap, extreme responses dominate the averages and the estimated difference reverses sign. Unlike the text and parking experiments, this comparison does not randomly assign party cues.

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SI 1 Treatments

Figure SI 1.1: Text treatment

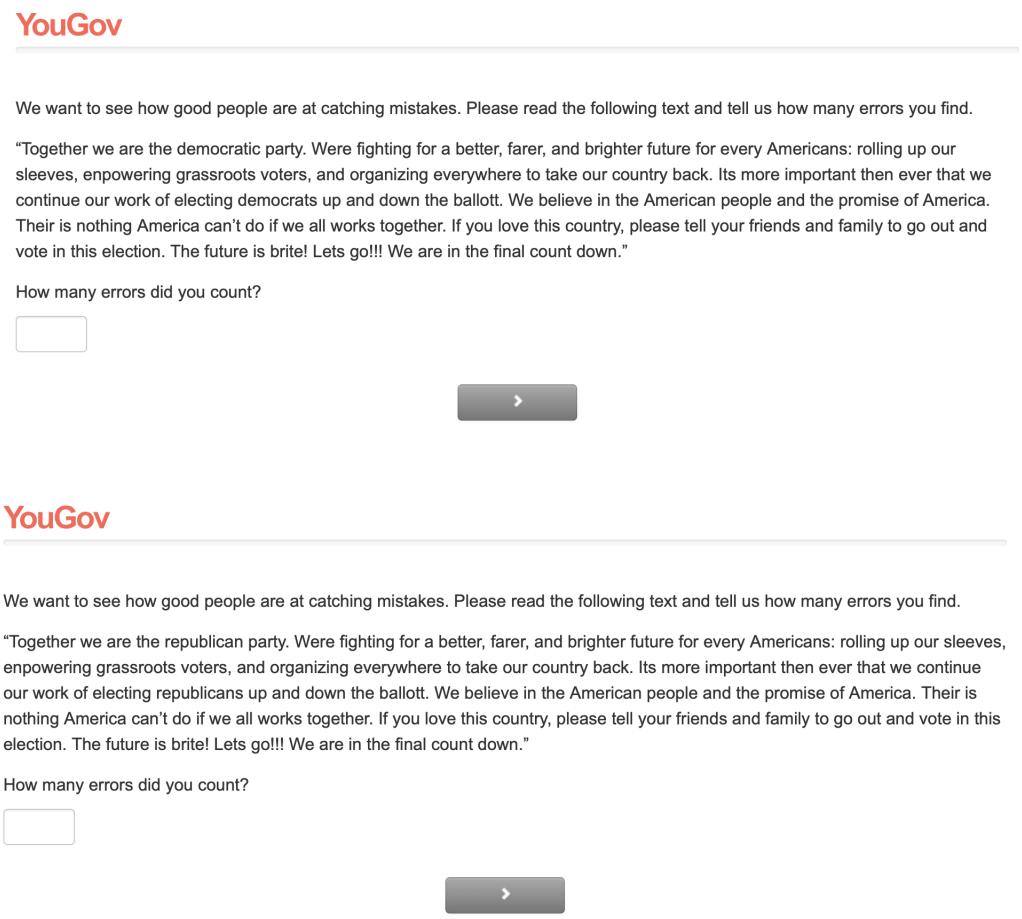
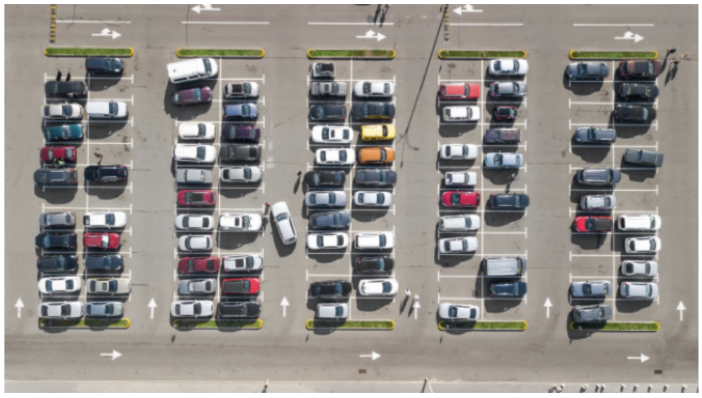


Figure SI 1.2: Parking lot

YouGov



We want to see how much attention people are paying in answering questions. How many cars would you say are poorly parked in this picture of the parking lot outside a Democratic Party meeting last year?

SI 2 Descriptive statistics and experimental contrasts

Tables SI 2.1–SI 2.4 report the descriptive statistics, including independents. Independents do not have an in-party or out-party cue; their means are reported by the party named in the treatment. Table SI 2.5 reports the experimental contrasts for Democrats and Republicans.

Table SI 2.1: CCES Writing

Respondents	Attributed party	N	Mean	Median	SE
Democrat	Democrat	334	8.65	9.00	0.23
Democrat	Republican	324	8.92	9.00	0.24
Independent	Democrat	94	9.07	9.00	0.72
Independent	Republican	110	8.24	8.00	0.43
Republican	Democrat	252	7.59	7.00	0.34
Republican	Republican	252	7.07	7.00	0.27

Table SI 2.2: Lucid Writing

Respondents	Attributed party	N	Mean	Median	SE
Democrat	Democrat	177	5.47	5.00	0.61
Democrat	Republican	132	5.91	5.00	0.78
Independent	Democrat	73	4.56	4.00	0.30
Independent	Republican	72	5.33	5.00	0.33
Republican	Democrat	111	5.58	4.00	0.89
Republican	Republican	109	4.86	5.00	0.26

Table SI 2.3: CCES Parking

Respondents	Attributed party	N	Mean	Median	SE
Democrat	Democrat	211	6.09	5.00	0.42
Democrat	Republican	212	8.72	6.00	0.60
Independent	Democrat	57	10.88	7.00	1.48
Independent	Republican	61	6.67	5.00	0.92
Republican	Democrat	181	8.43	6.00	0.63
Republican	Republican	163	8.09	5.00	0.71

Table SI 2.4: Mask counts capped at 15; never or rarely insincere

Respondents	N	Mean	Median	SE
Democrat	682	3.48	2.00	0.14
Independent	96	3.51	3.00	0.30
Republican	514	4.35	3.00	0.19

Table SI 2.5: Effect of party cues on error counts. Differences are out-party minus in-party means within respondent party. Intervals are pointwise Welch 95% confidence intervals.

Study	Task	Respondents	Difference	95% CI
CCES	Parking	Democrat	2.63	[1.19, 4.06]
CCES	Parking	Republican	0.34	[-1.53, 2.21]
CCES	Writing	Democrat	0.26	[-0.39, 0.92]
CCES	Writing	Republican	0.52	[-0.33, 1.37]
Lucid	Writing	Democrat	0.43	[-1.52, 2.39]
Lucid	Writing	Republican	0.71	[-1.13, 2.56]